

Museum exhibit clue cards (predict first)

Each card describes a real light you might meet in a museum or on the street. Predict the spectrum you would see through the grating and name the source type BEFORE you check it against the reference cards. This backs the exhibit-connection part of the score. Reasons are on the instructor answer key.

1 The OPEN sign

A shop's glowing orange-red OPEN sign in the window.

Spectrum I would see: _____ Source type: _____

2 The film projector

The warm bulb inside an old cinema film projector.

Spectrum I would see: _____ Source type: _____

3 The phone light

The white flashlight on the back of a modern phone.

Spectrum I would see: _____ Source type: _____

4 The street lamp

A highway lamp that makes everything look yellow-orange.

Spectrum I would see: _____ Source type: _____

5 The red firework

A single red burst at the fireworks show.

Spectrum I would see: _____ Source type: _____

6 The desk lamp

A dimmable desk lamp with an old-style clear bulb.

Spectrum I would see: _____ Source type: _____

Instructor answer key (do not hand out)

Keep this page with the instructor. A hot filament gives a smooth rainbow; a white LED gives a broad band with a blue spike; a neon, sodium, or metal-salt source gives separate bright lines.

#	SPECTRUM	SOURCE
1	Line spectrum	Neon gas
2	Smooth rainbow	Hot filament
3	Broad band with a blue spike	White LED
4	Line spectrum	Sodium gas
5	Line spectrum	Glowing strontium salt
6	Smooth rainbow	Hot filament